
Edge of Stability in Diffusion Models

Viola Tan

viola_tan@gsd.harvard.edu

Arthur De Los Santos

adls@mit.edu

Abstract

Adaptive gradient methods like Adam have been found to train at the *Adaptive Edge of Stability* (AEoS), where the largest eigenvalue of the preconditioned Hessian equilibrates at a stability threshold determined by step size η and Adam parameter β_1 . While this phenomenon is well-documented in standard supervised learning, its role in diffusion models—which involve a multi-task optimization across varying noise levels (σ)—remains underexplored. This study examines current stabilization techniques in diffusion models, such as EDM preconditioning, v -parameterization, and Min-SNR Weighting, through the lens of EoS. We argue that insights from σ -stratified curvature measurements can transform into principled diagnostics for stable diffusion training.

1 Introduction

The *Edge of Stability* (EoS) is the phenomenon where gradient descent operates with sharpness, the largest eigenvalue of the loss Hessian, hovering near or slightly above the stability threshold ($\approx 2/\eta$). In this regime, loss may fluctuate non-monotonically in the short term yet improve steadily over time [2]. This behavior has been shown to carry over to adaptive gradient methods like Adam, where the preconditioned Hessian stabilizes at the *Adaptive Edge of Stability* (AEoS) [4]. However, little is known about the sharpness dynamics in diffusion models, whose Adam-based training [8] effectively constitutes a multi-task optimization problem across a continuum of noise levels.

2 Related Works

2.1 Sharpness behavior in optimizers

Edge of Stability behavior has been shown to exist across optimizers, with the stability threshold being around $2/\eta$ for full-batch, vanilla Gradient Descent (GD) and Stochastic Gradient Descent (SGD) [2]. GD can be accelerated by the use of momentum; with the Heavy Ball (HB) momentum, the thresholds are $\frac{2+2\beta_1}{\eta}$ for StandardHB(η, β_1), and $\frac{2+2\beta_1}{\eta(1-\beta_1)}$ for EMAHB(η, β_1) [4]. Preconditioned Gradient Descent will diverge if the maximum eigenvalue of the preconditioned Hessian $P^{-1/2}HP^{-1/2}$ (the preconditioned sharpness) exceeds the stability threshold of the non-preconditioned algorithm. Adaptive gradient methods are variants of preconditioned gradient descent in which the preconditioner is changing rather than fixed. RMSPROP(η, β_2, ϵ) [16] and ADAM($\eta, \beta_1, \beta_2, \epsilon$) [12] are examples of adaptive gradient methods. Cohen et al. [4] show that adaptive methods train at the *Adaptive Edge of Stability* (AEoS), and their behavior differs significantly from that of non-adaptive methods at the *Edge of Stability* (EoS): while non-adaptive algorithms at the EoS are blocked from entering high-curvature regions of the loss landscape, adaptive gradient methods at the AEoS keep advancing into high-curvature regions, while adapting the preconditioner to compensate.

2.2 Diffusion models

2.2.1 Noise-Level Dynamics and the Multi-Task View

Diffusion models are trained by denoising Gaussian-corrupted inputs at varying noise levels σ_ϵ (or timesteps t), optimizing a denoising score-matching objective to predict either the injected noise ϵ , the clean signal x_0 , or a stabilized blend v . Each σ defines a distinct denoising subtask, creating a large-scale multi-task optimization where subtask hardness and gradient magnitudes vary sharply with σ . While studies examined the loss landscape in diffusion model training and suggested that low-noise regimes (small σ) are brittle and high-noise regimes (large σ) are smooth and easy [17], exact curvature measures and its relation to EoS remain underexplored.

2.2.2 Preconditioning and Scale Equalization

A classic stabilization strategy, demonstrated by *Elucidating the Design Space of Diffusion Models* (EDM), introduces σ -dependent preconditioning via multipliers on inputs, skip connections, and outputs ($c_{in}, c_{skip}, c_{out}$), along with matched loss weights $w(\sigma)$ and sampling distributions $p(\sigma)$. These maintain near-unit activations and targets across σ , producing more uniform gradients and robust training improvements [9].

EDM also clarifies target parameterizations. The v -parameterization [14] reduces target variance across σ , moderating instabilities. A related analysis of ADM-style U-Nets identifies “magnitude drifts” across noise levels and introduces normalization and uncertainty-based weighting to balance learning rates across σ , improving FID and stability while observing complex interactions between EMA, guidance, and optimization dynamics [10].

2.2.3 Reweighting and Task Interference

Viewing timesteps as tasks also motivates loss reweighting to manage interference. The Min-SNR- γ method defines a signal-to-noise ratio (SNR) per timestep and clamps extreme values so that neither low- nor high-SNR regions dominate optimization. This prevents a few σ values from driving parameter updates and reduces cross-task conflicts, accelerating convergence and improving final quality [7].

A complementary line quantifies “task affinity” between timesteps and finds that large σ gaps produce negative transfer; clustering timesteps (e.g., by SNR) and applying gradient balancing within clusters restores performance [6].

2.2.4 Adaptive Sampling and Target Stabilization

Beyond reweighting, adaptive samplers dynamically reallocate training effort toward impactful timesteps. By tracking per-timestep gradient variance and estimating each timestep’s expected objective reduction, these methods preferentially sample σ values predicted to yield the most progress. This accelerates convergence and produces interpretable σ -wise gradient statistics for monitoring [11].

Separately, Stable Target Field (STF) reduces supervision variance by averaging targets with a reference batch, particularly at intermediate σ where multimodality increases noise. This smooths optimization and complements EDM-like architectures without altering schedules [18].

2.3 Theoretical Perspectives on Stability

Emerging theory examines “score stability,” an analogue of algorithmic stability, to formalize generalization gaps under stochastic gradient descent for denoising score matching. These analyses incorporate learning-rate decay, gradient clipping, and weight decay as implicit regularizers, clarifying the optimizer’s stabilizing role. However, they omit σ -specific curvature effects and adaptive optimizers such as Adam, and do not yet provide σ -stratified stability bounds [5].

2.4 Gaps in Existing Literature

Across this literature, the empirical consensus is that σ -aware design via EDM preconditioning, SNR-based loss weighting, v -parametrization, adaptive sampling, and STF smoothing makes diffusion

training faster and more robust across unconditional, class-conditional, and text-to-image tasks. Yet, no work directly measures σ -stratified curvature or Adam’s effective step sizes along sharp eigenmodes.

3 Methodology

To investigate the Edge of Stability (EoS) phenomenon in diffusion models, we devised an experimental framework that monitors the spectral properties of the loss landscape across the diffusion trajectory. Our setup integrates standard diffusion architectures with exact Hessian spectral analysis methods adapted for adaptive optimizers.

3.1 Model Architecture and Dataset

We utilized a U-Net architecture based on the design by Ho et al. [8], adapted for the CIFAR-10 dataset (32×32 resolution). The network consists of a contracting path and an expanding path with skip connections. The time conditioning is implemented via sinusoidal positional embeddings projected through a Multi-Layer Perceptron (MLP) and injected into Residual Blocks via scale-and-shift modulation (AdaGN).

Self-attention mechanisms are applied at the 8×8 resolution bottleneck using Flash Attention for computational efficiency. The model width is controlled by a base channel dimension of 32, with channel multipliers of $[1, 2, 4]$ at subsequent downsampling stages. The total parameter count is approximately 2.2M.

Training was performed on the CIFAR-10 dataset, normalized to $[-1, 1]$. Due to the extreme computational cost of computing full-batch Hessians and gradients at every step, we restricted the training set to a random subset of 10,000 images (out of the standard 50,000 training images). This subset size serves as a practical "full-batch" setting for our experiments, allowing us to observe deterministic gradient descent dynamics while fitting within computational budgets.

3.2 Optimization and Hyperparameters

All models were trained using the Adam optimizer [12] with default hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. We explored a range of learning rates $\eta \in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}, \dots\}$ to analyze stability thresholds. Training was conducted for 6 hours (less than 1,000 epochs). Crucially, we trained using full-batch gradient descent on our 10,000-image subset (Batch Size $B = 10,000$). This eliminates gradient noise from mini-batching, ensuring that the observed training dynamics are governed purely by the deterministic Adaptive Edge of Stability (AEoS) mechanism rather than stochastic noise.

3.3 Curvature and Stability Measurement

The central metric of our analysis is the sharpness of the loss landscape, defined as the largest eigenvalue (λ_1) of the Hessian matrix H . Since Adam operates as a preconditioned gradient method, stability is governed by the eigenvalues of the preconditioned Hessian $P^{-1/2} H P^{-1/2}$ rather than the raw Hessian [4].

We implemented an online spectral analyzer that computes λ_1 at every training step. The preconditioner P is constructed dynamically from the Adam optimizer’s running second moment estimate ($\hat{v}_t$):

$$P_t = \text{diag} \left(\sqrt{\hat{v}_t} + \epsilon \right) \quad (1)$$

To compute the leading eigenvalue $\lambda_1(P_t^{-1/2} H_t P_t^{-1/2})$ efficiently without materializing the full Hessian, we utilized the Locally Optimal Block Preconditioned Conjugate Gradient (LOBPCG) algorithm [13, 1, 3]. This iterative method relies solely on Hessian-vector products (HVPs), which we computed using PyTorch’s automatic differentiation engine (`torch.autograd.grad`).

Estimation Approximation While training is performed on the full batch of 10,000 images, computing λ_1 on this entire set at every iteration is computationally prohibitive. Therefore, we

estimate the spectral norm using a random sub-sample of $B_{\text{measure}} = 1,000$ images. We argue that this sub-sample provides a sufficiently accurate approximation of the local loss landscape curvature to track EoS behavior. Since the optimization trajectory itself is driven by the full-batch gradient (10,000 images), the system remains in the deterministic AEoS regime; the estimator merely serves as a noisy, yet computationally feasible probe of this trajectory.

3.4 Timestep-Stratified Analysis

A key contribution of our methodology is the stratification of sharpness measurements by noise level. At regular intervals during training, we paused the weight updates to probe the local curvature at specific, fixed timesteps $t \in \{1, 100, 200, 300, 400, 500, 600, 700, 800, 999\}$. This distinguishes between the "global" sharpness (the curvature experienced by the optimizer at the randomly sampled training timestep) and the "local" sharpness intrinsic to specific noise levels σ_t .

3.5 Experimental Variants

We evaluated the curvature dynamics across six distinct diffusion implementations to isolate the effects of different stabilization strategies:

- **Baseline DDIM:** Standard ϵ -prediction with uniform timestep sampling [15].
- **EDM Preconditioning:** Continuous noise levels with scale-equalized input/output weighting [9].
- **V-Parameterization:** Predicting velocity $v_t \equiv \alpha_t \epsilon - \sigma_t x_0$ to avoid singularities at low SNR [14].
- **Min-SNR Weighting:** Reweighting the loss by $\min(\text{SNR}(t), \gamma)$ to clamp gradients from high-SNR regions [7].
- **Adaptive Sampling:** A learnable sampler parameterized by a Beta distribution, optimized via policy gradient to maximize loss reduction [11].
- **Stable Target Field (STF):** Reducing target variance by regressing towards a batch-averaged reference field [18].

4 Experiments and Results

4.1 Baseline DDIM

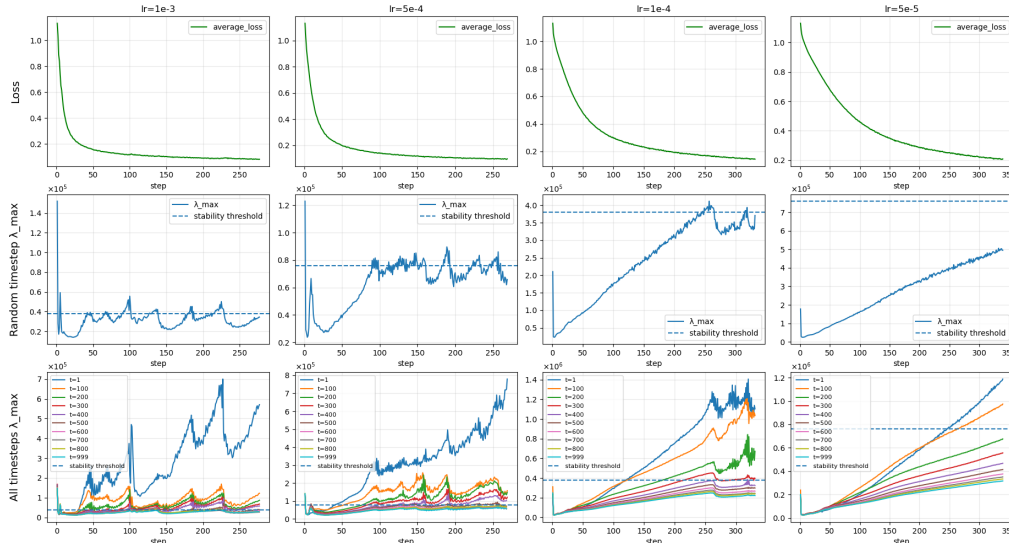


Figure 1: Baseline DDIM

During training, we measured the preconditioned sharpness $\lambda_1(P^{-1}H_t)$ both at the timestep randomly drawn for the weight update at each step, and at a collection of fixed diffusion timesteps $t \in \{1, 100, 200, \dots, 800, 999\}$. We varied the learning rate while holding the Adam hyperparameters fixed ($\beta_1 = 0.9$, $\beta_2 = 0.999$). For each experiment, we also plot the stability threshold of EMA-style heavy-ball momentum, $\frac{2+2\beta_1}{(1-\beta_1)\eta} = \frac{38}{\eta}$, following the analysis of the Adaptive Edge of Stability (AEoS).

Across learning rates for which training remains stable, the preconditioned sharpness at the *random* weight-updating timestep exhibits the expected AEoS behavior: it increases rapidly early in training and then equilibrates close to the theoretical stability boundary of $\frac{38}{\eta}$.

The trajectories for the fixed timesteps largely co-move with the random-timestep sharpness. When the random-timestep sharpness oscillate after reaching the stability threshold, we observe a similar oscillation for the fixed timesteps. However, the magnitudes of these λ_1 's differ systematically with the diffusion timestep. Earlier timesteps (small t , corresponding to cleaner images with higher signal-to-noise ratio) consistently attain the largest preconditioned sharpness, while later, noisier timesteps yield substantially smaller values. This ordering is mostly stable across learning rates and over the course of training. We also observe that at learning rates 10^{-3} and 5×10^{-4} , once the random-timestep sharpness has stabilized, sharpness at all fixed timesteps plateaus as well, except for the $t=1$ timestep which continues to elevate. Intuitively, the loss landscape at early timesteps is more curved: the model must fit high-frequency details of relatively clean images, which leads to larger eigenvalues of the (preconditioned) Hessian. Notably, unlike all other timesteps where sharpness seems to reach equilibrium in the $\eta = 10^{-3}$ training, $t = 1$ does not reach equilibrium within the available training budget.

4.2 EDM-Style Preconditioning

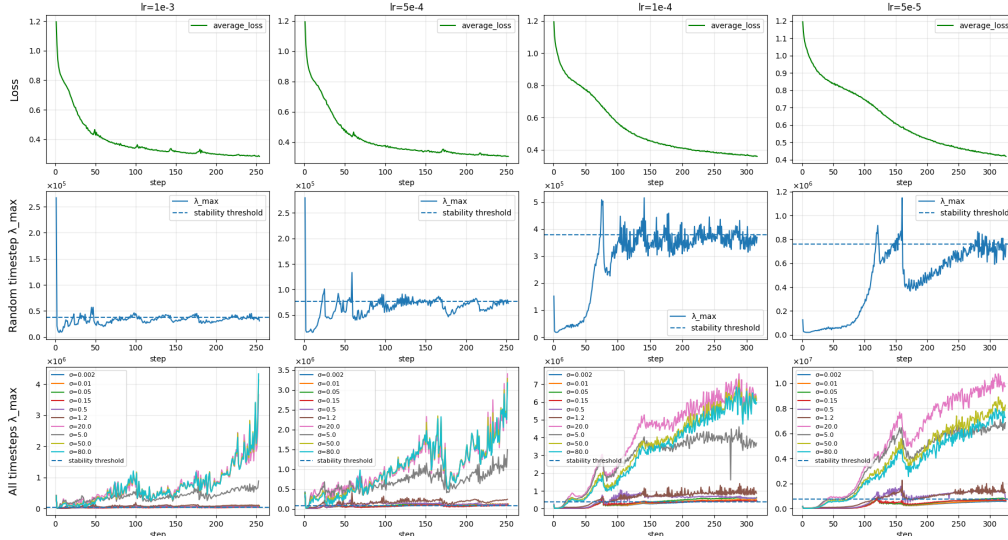


Figure 2: EDM-Style Preconditioning

In EDM parameterization [9] which predicts the clean image x instead of noise ϵ , we observe that higher noise levels consistently exhibit much larger preconditioned sharpness than lower noise levels, and for $\sigma \approx 20$ –80 the sharpness continues to grow throughout training at the largest learning rate ($\eta = 10^{-3}$), even after the sharpness at a randomly sampled σ has largely stabilized. This might suggest that the effective curvature in EDM is dominated by high-noise, low-SNR regions of the trajectory, which is the opposite of that in the base DDIM.

4.3 V-Parameterization

In the v -parameterization [14] experiments, the sharpness dynamics across diffusion timesteps differ substantially from those observed under ϵ -prediction: both the cleanest step ($t = 1$) and the highest-

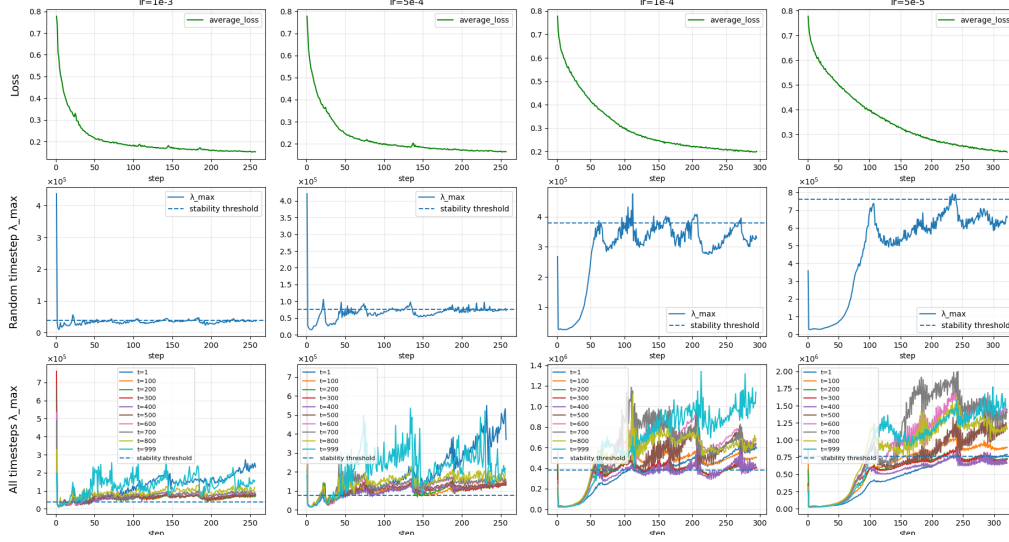


Figure 3: V-Parameterization

noise steps ($t \approx 700\text{--}999$) develop large preconditioned sharpness, while the mid-range timesteps ($t \approx 100\text{--}300$) remain comparatively flat. This produces a characteristic U-shaped structure in sharpness across the diffusion time axis. This behavior follows naturally from the definition of

$$v = \alpha_t \epsilon - \sigma_t x_0,$$

which produces a target that is a well-conditioned, SNR-normalized hybrid of the noise and clean image. Under ϵ -prediction, late timesteps suffer from extremely low SNR and consequently yield nearly uninformative and numerically unstable gradients. In contrast, the v -parameterization allows high-noise steps to retain a meaningful learning signal, increasing their induced curvature.

4.4 Min-SNR Weighting

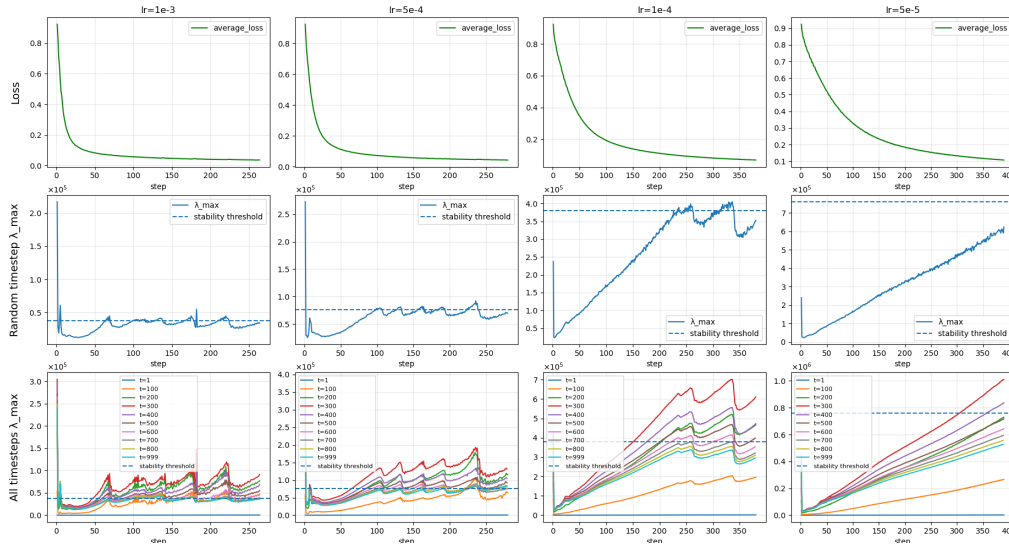


Figure 4: Min-SNR Reweighting

Min-SNR Weighting [7] prevents high SNR steps from dominating gradient update by clamping the weights applied to high-SNR steps and rescaling the loss so that low-SNR timesteps contribute more meaningfully. The resulting curvature dynamics reflect this rebalancing directly. Under Min-SNR

reweighting, the low-noise timesteps ($t \approx 1\text{--}100$) exhibit substantially suppressed preconditioned sharpness compared to the baseline settings. Because their loss contributions are explicitly down-weighted, these timesteps no longer dominate the local geometry of the loss landscape, and the largest eigenvalues of $P^{-1}H_t$ associated with these steps remain small throughout training. In contrast, the mid- and high-noise timesteps develop slightly larger sharpness values than the baseline, indicating that the reweighting increases their influence on the optimization dynamics.

4.5 Adaptive Sampling

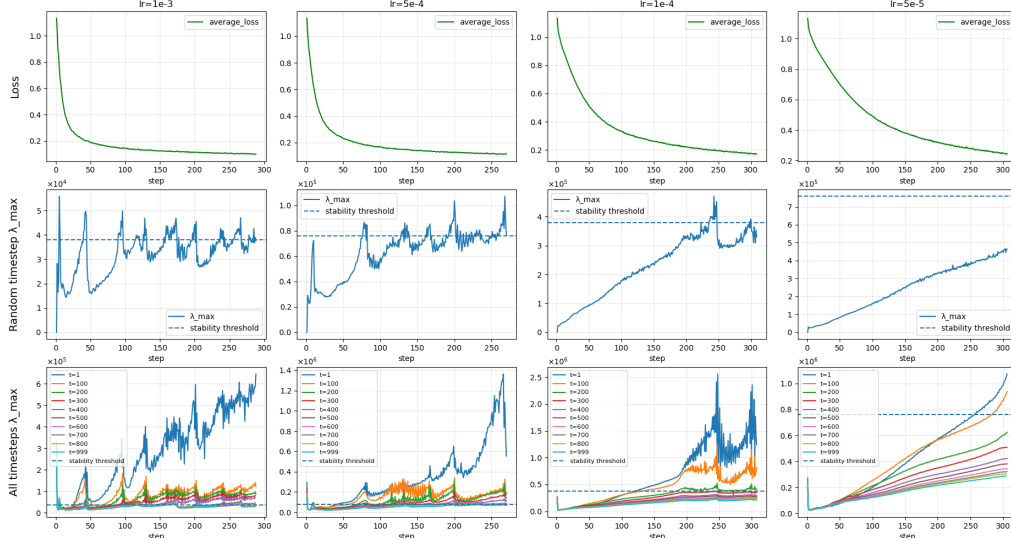


Figure 5: Adaptive Sampling

In terms of loss, adaptive sampling [11] achieves a very similar overall learning curve to the base model across all learning rates, but the sharpness dynamics are slightly different. At a higher learning rate ($\eta = 10^{-3}$), adaptive timestep sampling clearly stabilizes training: the preconditioned sharpness stays closer to the AEoS stability threshold and shows fewer large spikes than base DDIM. In contrast, at $\eta = 10^{-4}$ the adaptive schedule actually introduces more visible oscillations in sharpness than the baseline. A plausible explanation is that at large step sizes training is curvature-limited, and reweighting toward “easier” timesteps with moderate curvature damps instabilities and keeps Adam near its AEoS equilibrium. But once the learning rate is small enough that stability is no longer the bottleneck, the same non-uniform sampling mainly changes which timesteps dominate the gradient from step to step, effectively increasing gradient/Hessian variance and showing up as sharper oscillations in sharpness.

4.6 STF Smoothing

STF Smoothing replaces the regression target with a weighted combination of clean images drawn from a reference batch, effectively averaging over nearby targets. Its impact is most pronounced in the high-learning-rate experiment ($\eta = 10^{-3}$), where it substantially damps the oscillations in preconditioned sharpness, suggesting that the smoothed targets reduce curvature and gradient variance in this more unstable regime. For the smaller learning rates, where training is already relatively stable, STF Smoothing leaves both the loss and sharpness trajectories largely unchanged.

5 Discussion

Our core findings include:

AEoS behavior in random, weight-updating timestep. We found that in diffusion model training, the preconditioned sharpness of the random timestep that updates the weights at every step stabilizes

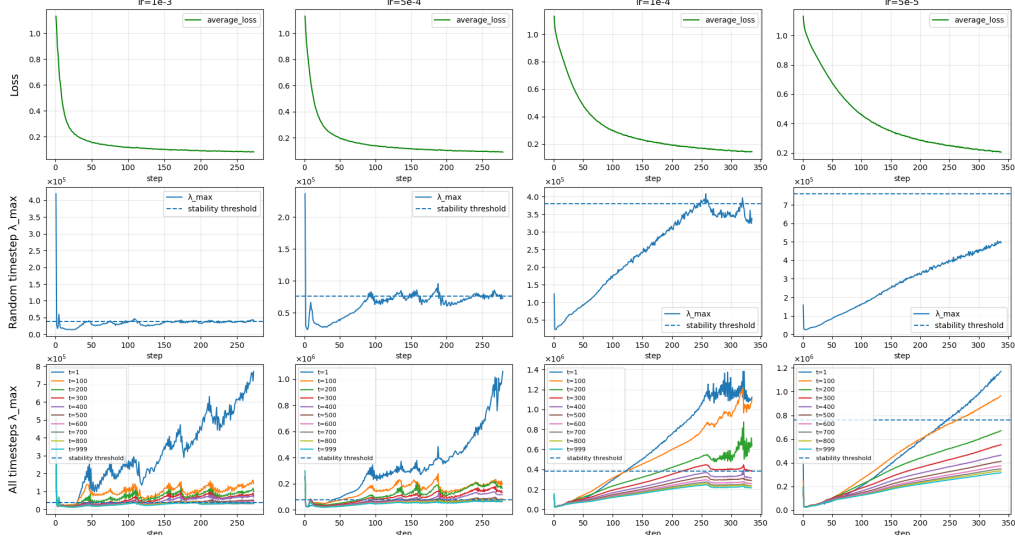


Figure 6: STF Smoothing

at the stability threshold of $\frac{2+2\beta_1}{(1-\beta_1)\eta} = \frac{38}{\eta}$, consistent with Adam behavior in standard supervised machine learning [4].

Dominance of random, weight-updating timestep. We also found that the sharpness of the random timestep largely dominates the dynamics of sharpness in other fixed timesteps. When the random stabilizes, a large portion of the fixed timestep sharpness also stabilizes, with the exception of some timesteps that continues to sharpen, and this group of timesteps vary by the diffusion techniques that we employ.

Diffusion techniques affect loss landscapes across timesteps. In the original DDIM, the magnitude of the sharpness remains organized by timesteps, where the smallest timesteps (largest SNR) mostly have the largest sharpness throughout training. However, different diffusion techniques have different effects on this order. Techniques like EDM preconditioning and v-parameterization that change the diffusion target have a significant impact on this order: in EDM, the noisiest σ 's have the largest sharpness; in v-parameterization, the timesteps with the lowest and highest noise have the largest sharpness. Min-SNR suppresses the preconditioned sharpness of the low-noise timesteps. Adaptive sampling and STF smoothing have relatively small impact on the sharpness behavior.

6 Conclusion

Our findings reveal that diffusion model training exhibits a characteristic AEoS-like stabilization dynamic governed primarily by the random, weight-updating timestep, whose preconditioned sharpness converges to the Adam stability threshold and in turn shapes the sharpness behavior of most fixed timesteps. While the majority of timesteps inherit this stabilization, a subset continues to sharpen in ways that depend systematically on the diffusion technique employed. This highlights that diffusion parameterizations and preprocessing choices such as DDIM, EDM, v-parameterization, Min-SNR, adaptive sampling, and STF smoothing meaningfully reshape the loss landscape across timesteps, sometimes inverting or suppressing expected sharpness hierarchies. Overall, these results suggest that the interplay between timestep-specific sharpness and diffusion parameterization is central to understanding optimization stability and generalization behavior in diffusion models.

7 Limitations

Our study faces several limitations primarily driven by computational constraints. First, training was restricted to a fixed budget of approximately 6 hours per experiment. Consequently, we were

unable to reach the target of 1,000 epochs, and the models did not reach full convergence. While we observed the characteristic rapid rise and stabilization of sharpness typical of the early-to-mid training phase, we could not observe the terminal dynamics of the loss landscape, such as whether the $t = 1$ timestep sharpness eventually stabilizes upon full convergence. Additionally, technical challenges with resetting optimizer states upon checkpoint loading prevented us from resuming training runs to extend this duration.

Second, to approximate full-batch gradient descent dynamics within memory limits, we restricted the training set to a random subset of 10,000 images from CIFAR-10, rather than the full 50,000. Similarly, the exact Hessian spectral analysis was estimated using a sub-sample of 1,000 images. While we argue that this subsample is sufficient to probe local curvature, the reduced dataset size may affect the generalization dynamics and the precision of the eigenvalue estimates compared to a calculation over the full data distribution.

Third, due to the lack of convergence, we did not report Fréchet Inception Distance (FID) scores. Qualitative inspection revealed that while baseline models remained noisy after 6 hours, stabilized variants like EDM produced recognizable (though object-indistinct) CIFAR-like structures, suggesting different rates of feature learning that our current curvature metrics do not fully capture.

To address these limitations, future work should prioritize training on the full dataset for 1,000+ epochs to correlate asymptotic curvature behavior with generation quality (FID). Furthermore, while our results are empirical, future research should aim to derive analytical bounds for the stability threshold in the specific context of multi-scale diffusion objectives, moving beyond the general AEoS theory.

8 Implications and Significance

This work bridges a critical gap between the theory of optimization stability and the practice of generative modeling. By confirming that diffusion models trained with Adam operate at the Adaptive Edge of Stability (AEoS), we provide a theoretical framework for understanding why certain heuristic modifications—such as noise schedules and parameterizations—succeed or fail.

Theoretical Significance. To the best of our knowledge, this is the first study to empirically map the σ -stratified sharpness dynamics of diffusion models. Our findings challenge the view of diffusion training as a monolithic optimization problem, revealing it instead as a delicate balancing act of coupled sub-tasks where distinct noise levels can drive the optimizer into instability. This extends the applicability of the Edge of Stability hypothesis beyond standard supervised learning into hierarchical, multi-task generative settings, suggesting that "riding the edge" of the stability threshold is a universal characteristic of effective deep learning optimization.

Practical Consequences. For practitioners, our results suggest that curvature monitoring can transform from a theoretical curiosity into a principled diagnostic tool. Currently, designing noise schedules or weighting functions (like Min-SNR) requires expensive trial-and-error. Our framework implies that these hyperparameters could instead be calibrated to align the spectral norms of different timesteps with the optimizer's stability threshold. For instance, the observation that EDM and v-parameterization invert the sharpness hierarchy suggests that "stability" is malleable; practitioners can engineer the loss landscape to suppress gradients from unstable modes (as Min-SNR does) or homogenize curvature across time.

Broader Impact. Beyond diffusion, these insights have implications for any domain involving multi-scale or curriculum-based learning. The dominance of specific "hard" tasks (like low-noise timesteps) in driving global sharpness highlights a general mechanism for task interference in multi-task learning. Understanding these dynamics paves the way for automated curricula that dynamically adjust task weights to maintain optimal sharpness, potentially reducing the energy and data requirements for training large-scale generative models.

Acknowledgments and Disclosure of Funding

The authors acknowledge the MIT Office of Research Computing and Data (ORCD) for providing high-performance computing resources on the Engaging Cluster that contributed to the results reported in this paper (ORCD Docs, 2025). We thank Prof. Tomaso Poggio, Prof. Lorenzo Andrea Rosasco, and Pierfrancesco Beneventano for their support! We would also like to thank

References

- [1] Arseniy Andreyev and Pierfrancesco Beneventano. Edge of stochastic stability: Revisiting the edge of stability for sgd, 2025.
- [2] Jeremy Cohen, Simran Kaur, Yuanzhi Li, J. Zico Kolter, and Ameet Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability. *CoRR*, abs/2103.00065, 2021.
- [3] Jeremy M. Cohen, Alex Damian, Ameet Talwalkar, J. Zico Kolter, and Jason D. Lee. Understanding optimization in deep learning with central flows, 2025.
- [4] Jeremy M. Cohen, Behrooz Ghorbani, Shankar Krishnan, Naman Agarwal, Sourabh Medapati, Michal Badura, Daniel Suo, David Cardoze, Zachary Nado, George E. Dahl, and Justin Gilmer. Adaptive gradient methods at the edge of stability, 2024.
- [5] Tarek Farghly, Patrick Rebeschini, George Deligiannidis, and Arnaud Doucet. Implicit regularisation in diffusion models: An algorithm-dependent generalisation analysis. 2025.
- [6] Hoki Go, Jonghyun Kim, Yunhui Lee, Seungjun Lee, Sewoong Oh, Heung-Yeung Moon, and Sungjin Choi. Addressing negative transfer in diffusion models. 2023.
- [7] Tian Hang, Shuhang Gu, Chao Li, Jianmin Bao, Dong Chen, Han Hu, Xiangyu Geng, and Baining Guo. Efficient diffusion training via min-snr weighting strategy. 2023.
- [8] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models, 2020.
- [9] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. 2022.
- [10] Tero Karras, Miika Aittala, Jaakko Lehtinen, Janne Hellsten, Timo Aila, and Samuli Laine. Analyzing and improving the training dynamics of diffusion models. 2023.
- [11] Minjun Kim, Donghyeon Ki, Seung-Woo Shim, and Byung-Jun Lee. Adaptive non-uniform timestep sampling for diffusion model training. 2024.
- [12] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. Published in ICLR 2015.
- [13] Andrew V. Knyazev. Toward the optimal preconditioned eigensolver: Locally optimal block preconditioned conjugate gradient method. *SIAM Journal on Scientific Computing*, 23(2):517–541, 2001.
- [14] Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models. In *International Conference on Learning Representations (ICLR)*, 2022. Preprint version at arXiv.
- [15] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models, 2022.
- [16] Tijmen Tieleman and Geoffrey Hinton. Lecture 6.5: Rmsprop: Divide the gradient by a running average of its recent magnitude. Coursera: Neural Networks for Machine Learning lecture slides, 2012. Online lecture, University of Toronto.
- [17] Tian Xu, Peiran Mi, Ruoyu Wang, and Yuxin Chen. Towards faster training of diffusion models: An inspiration of a consistency phenomenon. 2024.
- [18] Yilun Xu, Siyi Tong, and Tommi Jaakkola. Stable target field for reduced variance score estimation in diffusion models. 2023.

A Code Availability

The complete source code for this project, including the implementations of all six diffusion variants (Baseline, EDM, V-Param, Min-SNR, Adaptive Sampling, and STF) and the spectral analysis tools, is available at:

<https://github.com/arthurdls/Edge-of-Stability-on-Diffusion>

This repository contains:

- `implementations/`: PyTorch modules for each diffusion stabilization technique.
- `train.py`: The main training script supporting full-batch approximation.
- `diffusion_training.ipynb`: An interactive notebook for running experiments.
- `implementations/utils/`: The exact Hessian spectral analysis tools (LOBPCG).

B Generated Sample Progression

This appendix presents the progression of generated samples throughout the training process for each of the six experimental variants. For every test, we visualize the model outputs across four learning rates: $\eta \in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}, 5 \times 10^{-5}\}$.

Each row displays samples taken at epochs 50, 100, 150, 200, and 250. Each individual image within the row consists of 16 generated samples arranged in a 4×4 grid, allowing for a qualitative assessment of sample diversity and quality.

B.1 Baseline DDIM

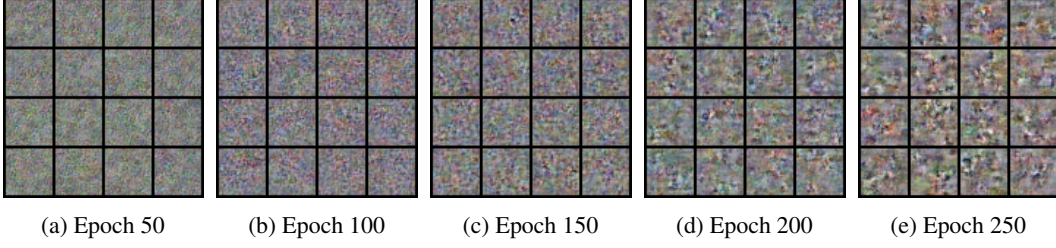


Figure 7: **Baseline DDIM** ($\eta = 10^{-3}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

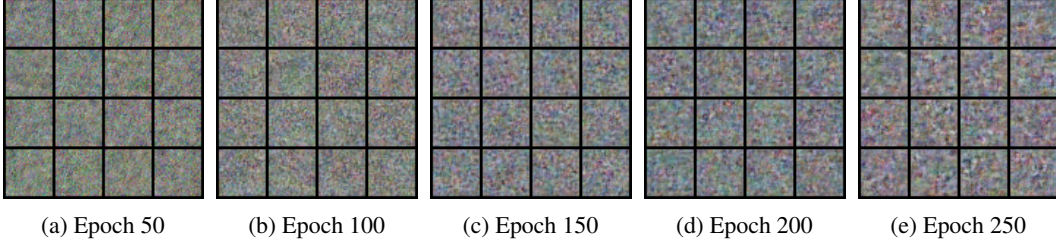


Figure 8: **Baseline DDIM** ($\eta = 5 \times 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

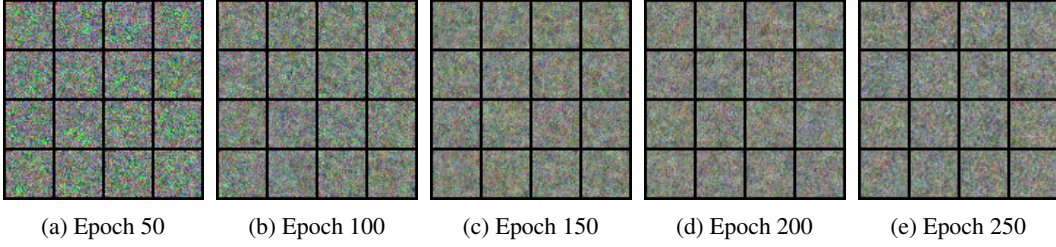


Figure 9: **Baseline DDIM** ($\eta = 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

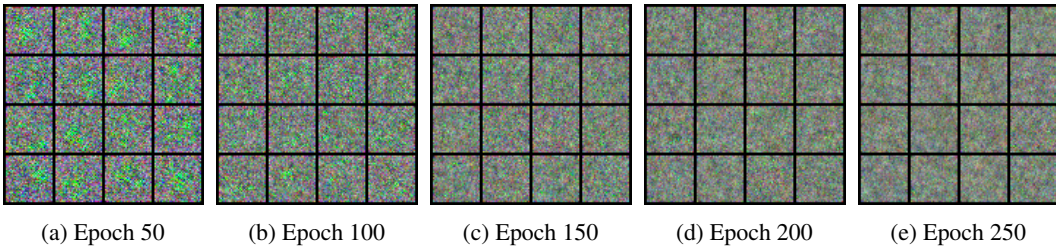


Figure 10: **Baseline DDIM** ($\eta = 5 \times 10^{-5}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

B.2 EDM Preconditioning

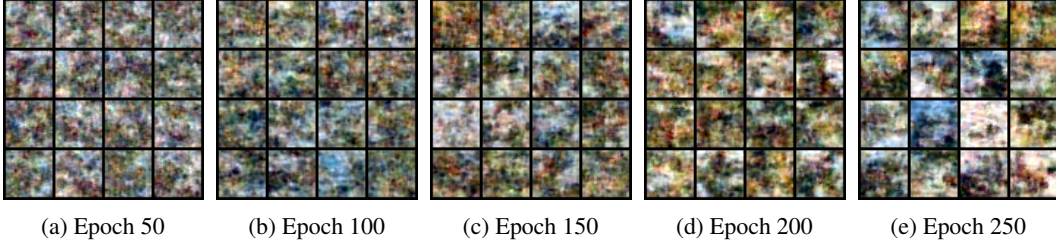


Figure 11: **EDM Preconditioning** ($\eta = 10^{-3}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

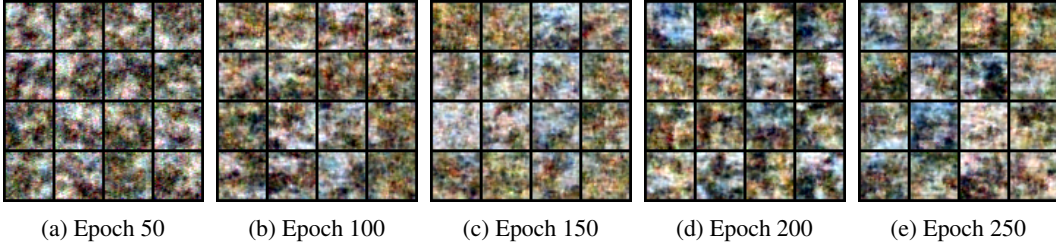


Figure 12: **EDM Preconditioning** ($\eta = 5 \times 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

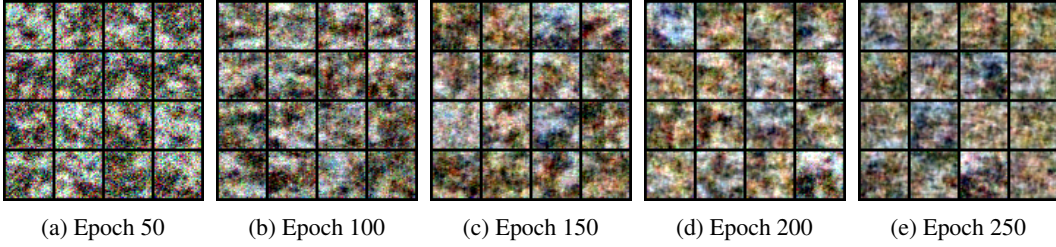


Figure 13: **EDM Preconditioning** ($\eta = 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

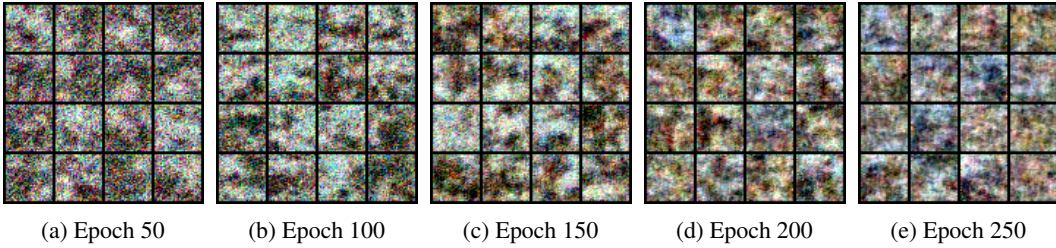


Figure 14: **EDM Preconditioning** ($\eta = 5 \times 10^{-5}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

B.3 V-Parameterization

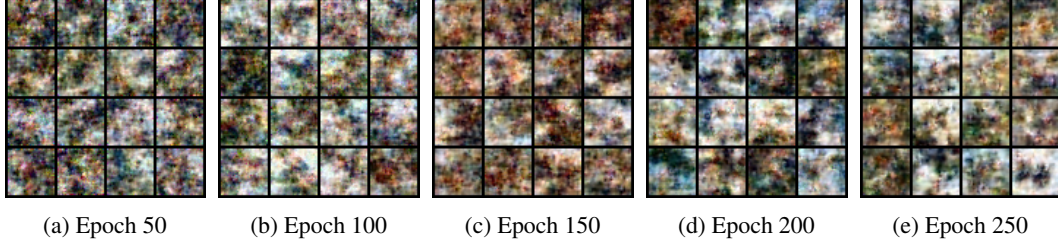


Figure 15: **V-Parameterization** ($\eta = 10^{-3}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

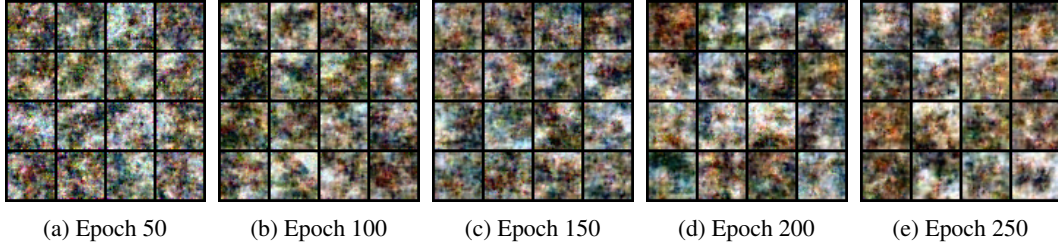


Figure 16: **V-Parameterization** ($\eta = 5 \times 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

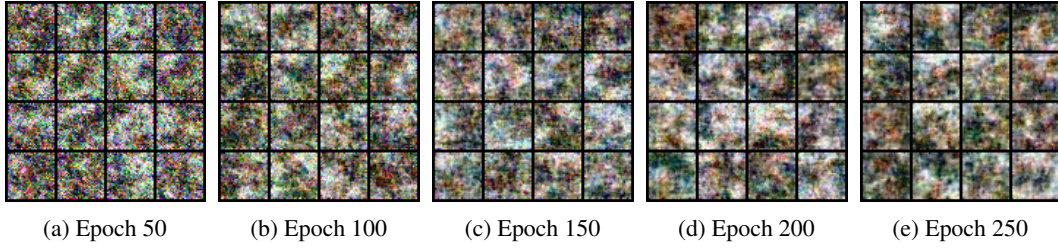


Figure 17: **V-Parameterization** ($\eta = 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

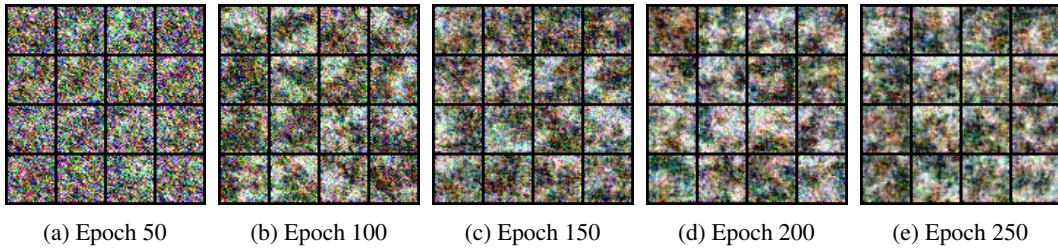


Figure 18: **V-Parameterization** ($\eta = 5 \times 10^{-5}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

B.4 Min-SNR Weighting

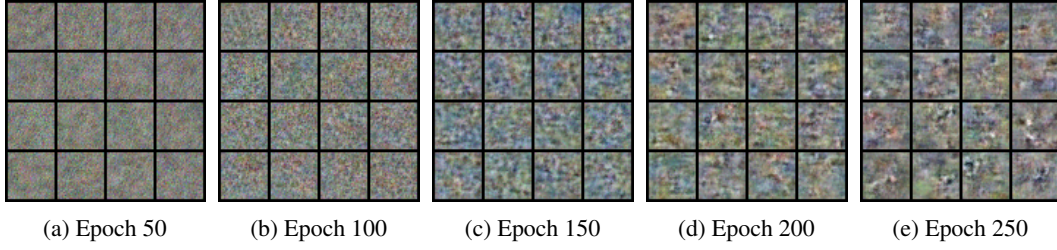


Figure 19: **Min-SNR Weighting** ($\eta = 10^{-3}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

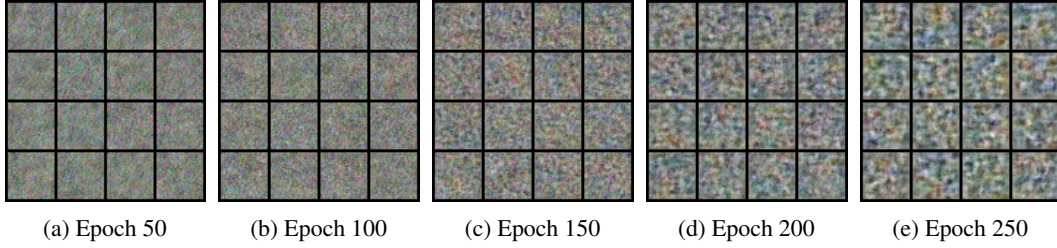


Figure 20: **Min-SNR Weighting** ($\eta = 5 \times 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

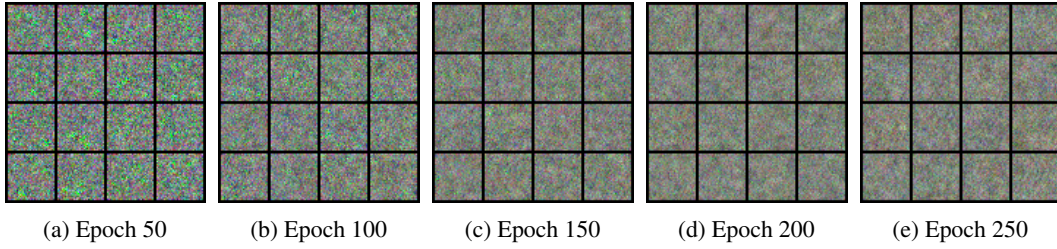


Figure 21: **Min-SNR Weighting** ($\eta = 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

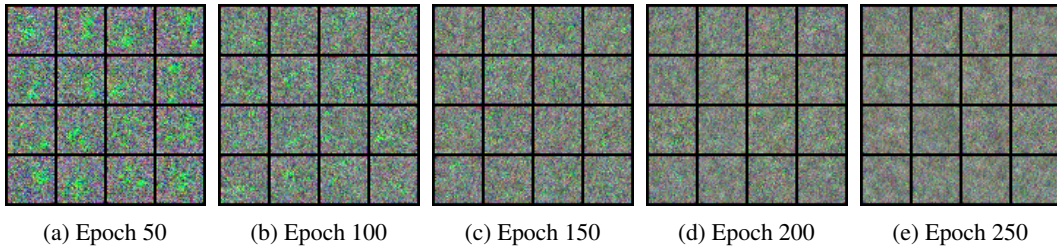


Figure 22: **Min-SNR Weighting** ($\eta = 5 \times 10^{-5}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

B.5 Adaptive Sampling

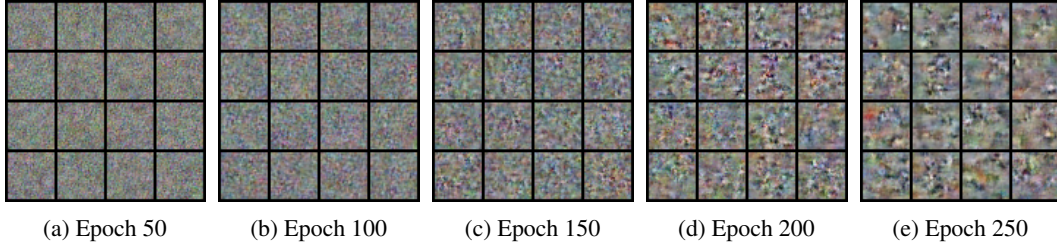


Figure 23: **Adaptive Sampling** ($\eta = 10^{-3}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

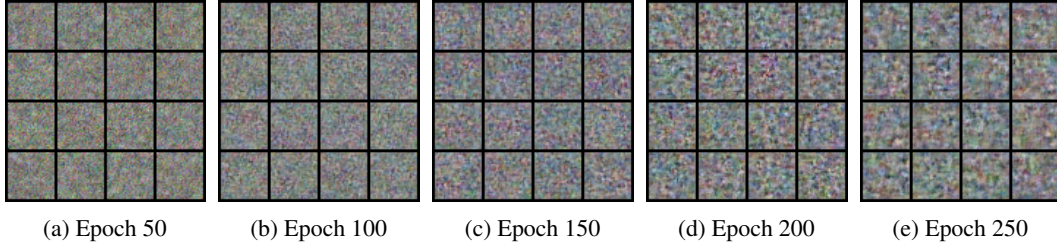


Figure 24: **Adaptive Sampling** ($\eta = 5 \times 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

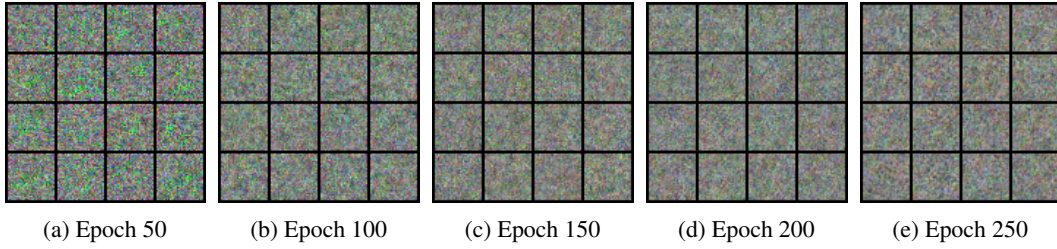


Figure 25: **Adaptive Sampling** ($\eta = 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

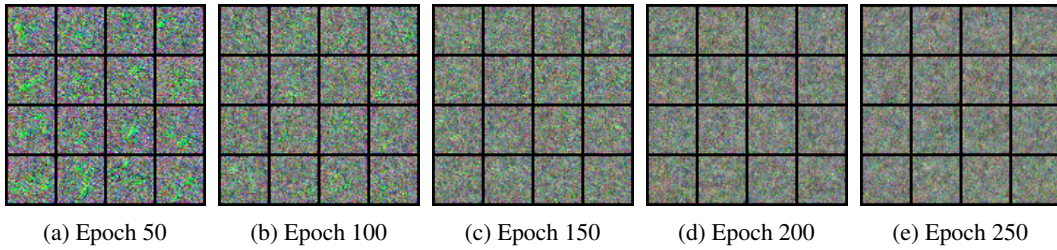


Figure 26: **Adaptive Sampling** ($\eta = 5 \times 10^{-5}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

B.6 STF Smoothing

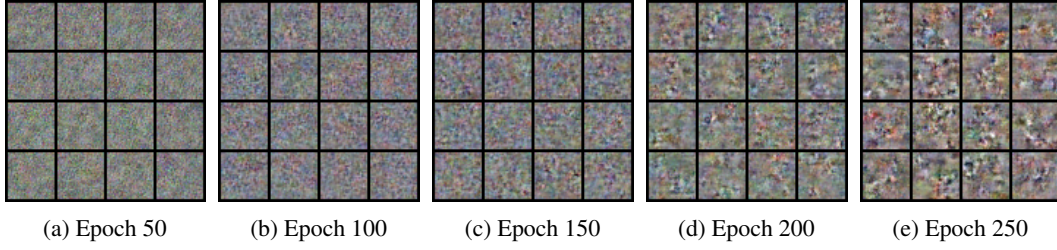


Figure 27: **STF Smoothing** ($\eta = 10^{-3}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

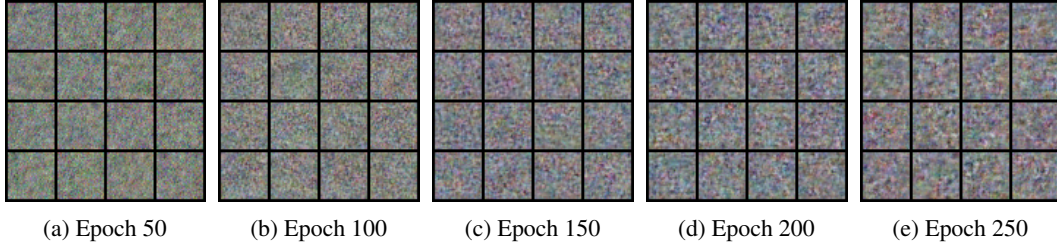


Figure 28: **STF Smoothing** ($\eta = 5 \times 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

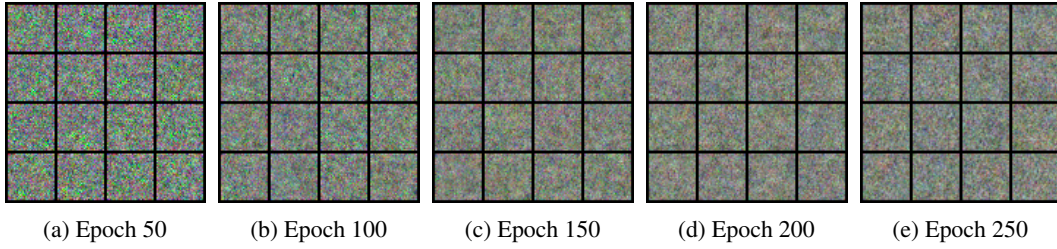


Figure 29: **STF Smoothing** ($\eta = 10^{-4}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.

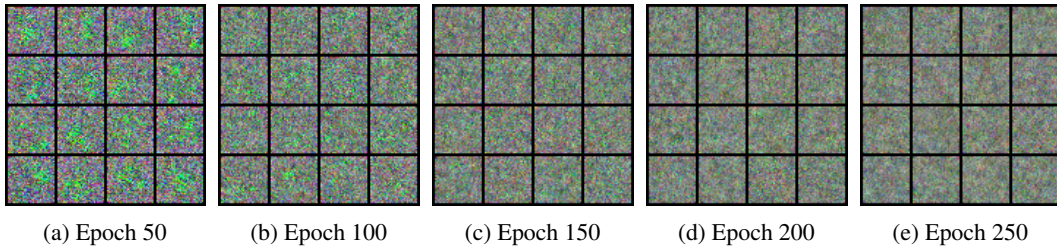


Figure 30: **STF Smoothing** ($\eta = 5 \times 10^{-5}$). Generated samples at varying training stages. Each panel displays a 4×4 grid of 16 images.